

Certainty Is Outsourced Trust: Mechanism-Grounded, Black-Box Fuzzing of Multi-Agent LLM Systems

(draft for review)

Abstract

Multi-agent LLM systems (MAS) increasingly authorize consequential tool actions behind a safety assumption: an upstream auditor agent will catch a malicious request before a decider acts. A recent measurement study—the *capability paradox*—unsettles this: a *more* capable auditor can make the system *more* hijackable, mediated by the auditor’s linguistic *certainty*. We turn that mediator into a fuzzing fitness. MASFUZZER is a black-box, coverage-guided fuzzer: an attacker LLM rewrites the entry request to climb the auditor’s output certainty while preserving a coherent incident narrative. In the canonical Manager–Worker topology, certainty-steering roughly *doubles* attack success over a fair same-LLM baseline (paired difference +0.176, cluster-bootstrap 95% CI [+0.09, +0.26] over 3 seeds, $n=216/\text{arm}$) with coherence held, and shows a certainty→hijack dose-response that survives a pseudoreplication-corrected (cluster-robust) test ($p<.01$) and replicates across three independent campaigns ($\rho \approx 0.25$ each). Sweeping five architectures, we find the advantage is *not* a universal rule but is governed by a single mechanism. Each topology is hijacked through whether the upstream content the decider trusts *endorses* the action—certainty for an auditor, endorsement-transfer for a handoff swarm, vote-balance for a broadcast group-chat—and the effect size scales with how much the decider *outsources* safety judgment to a separable upstream verdict. We confirm this causally: inserting an auditor into an otherwise-robust swarm raises ASR $0.36 \rightarrow 0.54$ and makes certainty a steerable causal lever ($\rho=0.26$, $p=.008$), even though the executor still judges. This resolves why the capability paradox’s *judging* manager is hijacked at all—it sits at the full-outsourcing end of the continuum.

1 Introduction

LLM multi-agent systems now triage incidents, write to databases, and call infrastructure tools with limited human oversight. A central safety assumption is that an upstream auditor or reviewer agent catches a malicious or mistaken request before a decider acts on it. The capability paradox [9] unsettles this: a *more* capable auditor can make the system *more* hijackable, because a confidently-worded “safe” verdict is precisely what persuades the decider to comply—a failure mode they term *semantic hijacking*, mediated by the auditor’s linguistic certainty.

Turning this observation into a systematic attack generator is unaddressed. White-box MAS fuzzers (FLARE [4]) need agent source and target functional reliability, not security; single-agent prompt-injection fuzzers (AGENTVIGIL [10]) do not model the multi-agent decision path; and quality-diversity red-teaming [2, 8, 11] searches input-side descriptors against a single model. None steers by the causal mediator of the failure, and none asks how the right attack depends on the MAS architecture.

We make the mediator the fuzzing fitness. MASFUZZER is black-box—it observes only agents’ text—and an attacker LLM rewrites the request to raise the auditor’s output certainty while pre-

serving a coherent incident narrative, hill-climbing this continuous, causally-grounded signal. We then ask where across architectures the lever pays off, and find a single unifying mechanism.

Contributions.

1. We turn the capability-paradox mediator into a fuzzing fitness and confirm, *in the attack loop*, a certainty→hijack dose-response that is robust to pseudoreplication (cluster-robust $p < .01$; consistent across three independent campaigns).
2. A black-box, mechanism-grounded LLM-mutation fuzzer whose certainty-steering roughly doubles ASR over a fair same-LLM baseline (0.45 vs 0.27) with coherence held.
3. From 500 payloads $\times$ 14 manager models we mine and *adversarially verify* the framings that make attacks succeed, and show coarse categories and input-text certainty do *not* predict success.
4. A mechanism for *which* signal hijacks *which* architecture: a single endorsement-transfer law modulated by judgment-*outsourcing*, confirmed causally, which resolves the capability-paradox puzzle and tells a defender exactly which topology to harden.

2 Threat Model and Problem

A MAS is an ordered set of agents $A_1, \dots, A_n$, each a fixed (model, system-prompt) pair connected by a topology τ . Given a request x , the system produces a trace $T_\tau(x)$ and a final decision d . A validated oracle $O(d) \in \{0, \dots, 4\}$ grades it; a *hijack* is $O(d) \geq 2$. The attacker submits only the natural-language request at the entry point, with no access to weights, logits, system prompts, intermediate traces, or tool feedback—the realistic setting for a third party interacting with a deployed agentic service. The objective is to maximize the hijack rate under a query budget B .

3 Method

Certainty as a climbable fitness. For the auditor (or first-agent) output o we score linguistic certainty $c(o)$ as the assertive-minus-hedging term density using the exact lexicon with which the mother paper’s mediation was measured. Certainty is a *continuous, climbable* scalar; behavioral-diversity descriptors, by contrast, saturate after a handful of samples and tie a random baseline.

Certainty-guided LLM mutation. An attacker LLM rewrites the parent request to maximize c while keeping one coherent incident narrative and preserving the dangerous action (few-shot on high-certainty exemplars). A $\mu + \lambda$ hill-climb breeds the top- k elites by c . The decisive baseline is the *same* LLM rewriting with no objective (NEUTRAL), isolating steering from LLM rewriting per se; we also compare to CONCAT, a hand-coded random-operator baseline.

Success-condition mining. Using 500 payloads $\times$ 14 real manager models from existing oracle-graded logs (zero new queries), a multi-agent pipeline contrasts the cross-manager success extremes, labels the corpus, and adversarially verifies each candidate framing against strategy/tool-matched controls.

4 Experiments and Results

Setup. We evaluate on `paradox_dataset_500` (500 SRE incident-response prompt-injection payloads). Weak agents are Llama-3.2-3B, the attacker/mutator and strong decider are DeepSeek-V3, and the oracle is the capability-paradox-validated GPT-4o-mini judge ($\kappa \geq 0.87$), under a strict decision SOP (the regime in which ASR is intermediate and guidance can discriminate). Primary

metric: ASR; secondary: deep-capture (grade ≥ 3), coherence (judge plausibility), dose-response. *Statistics.* We treat the random seed as the experimental unit (3 seeds, 72 candidates/arm/seed); the headline uses a cluster bootstrap (resampling candidates within each (seed,arm)), the dose-response a cluster-robust GEE plus per-seed correlations (to avoid pseudoreplication), and all mechanism p -values are Holm–Bonferroni corrected. We flag $n=3$ seeds as a limitation and recommend 5 for camera-ready.

Certainty is a climbable fitness. ASR climbs across worker-certainty bins from 0.20 to 0.52 (Fig. 2), with one honest attenuation at the extreme bin (maximally assertive reports begin to read as implausible). A naive pooled correlation ($\rho=0.25$, $p=5\times 10^{-11}$) over-states significance, since hill-climb candidates are autocorrelated; the effect nonetheless survives a cluster-robust GEE ($p=.010$) and replicates across all three independent campaigns ($\rho=0.25, 0.27, 0.26$; each $p<10^{-3}$). Crucially, the certainty of the *input payload* does *not* predict cross-manager success ($\rho \approx 0.05$, n.s.)—the lever is the auditor’s *output*.

Certainty-steering beats fair baselines. In the Manager–Worker topology at $n=216$ /arm (Fig. 3), certainty-steering reaches ASR 0.45 vs 0.27 (NEUTRAL) and 0.37 (CONCAT); the paired certainty–neutral difference is +0.176 with a cluster-bootstrap 95% CI of [+0.09, +0.26] (excludes 0; paired- t $p=.025$ over 3 seeds), and coherence is held (0.94 vs 0.88). The synthesized attacks read as realistic incident reports.

Cross-model robustness. The mechanism is not specific to one model pair. Replacing the weak agent with a second family (Qwen-2.5-7B), the certainty→hijack dose-response holds at the supervisor edge ($\rho=0.36\text{--}0.47$, $p<10^{-3}$) and the swarm+auditor causal anchor remains significant ($\rho(\text{auditor cert, grade})=0.31$, $p=.002$). The *more capable* Qwen worker is markedly *more* hijackable overall (certainty-arm ASR ceilings at 0.98–1.00 vs ≈ 0.45 for Llama-3.2-3B, and a strong DeepSeek decider does not lower it)—itself a prediction of the capability paradox (more capable auditor → more confident → more hijackable). The mechanism thus translates across the model axis, while its absolute strength scales with worker capability, as the paradox predicts.

What makes attacks succeed. Per-payload cross-manager success ranges 0.00–0.85. Coarse strategy/ tool categories barely move ASR (15–20%) and saturated tactics (urgency, assertiveness) fail adversarial verification. The verified conditions are framings of *how the destruction is presented*: a live external-attacker reframe (42–46% of high-success vs 8% of blocked payloads), a first-person commander voice with forensic certainty (67% vs 27%), a euphemized action verb (80% vs 8%), a named recovery source on an irreplaceable target, an omitted loss-disclosure, and third-party/compliance stakes.

Where does it generalize as an attack? Sweeping five architectures over four configurations (Fig. 4), exactly one cell yields a *stable, high-effect* certainty attack across all configurations and both scales: the supervisor (auditor→manager) topology ($0.52 \pm .06$). Every other topology is unstable at the ASR level (the single fused agent flips with scale; pipeline is a wash; group-chat and swarm are high-variance). We are careful to separate two questions: whether *steering a signal is a strong, reliable attack* (answered here—only at the supervisor edge), and whether *a signal is the causal mechanism of hijacking* in a topology (Sec. 5). A lever can be a genuine mechanism even where steering it is not yet a high-variance-free attack; the next section establishes the mechanism and anchors it with two causal experiments.

5 The Hijack Mechanism: an Outsourcing Continuum

Why does certainty dominate the supervisor but not other topologies? We probed each architecture for its own hijack lever, classifying every upstream agent’s stance toward the action with a judge

Table 1: Each architecture’s hijack lever (Spearman correlation with the hijack grade). All are the same underlying quantity—whether the upstream content the decider trusts *endorses* the action—in topology-native form. p -values are Holm–Bonferroni corrected over the seven lever/causal tests of Sec. 5; [†] marks cells that are Holm-significant but only *suggestive* under the stricter Bonferroni.

Architecture	Lever (form of endorsement)	ρ	raw p	Holm p
supervisor	auditor’s confident “safe” verdict (certainty)	+0.49	.0003	.002
swarm	triage/specialist endorsement (handoff)	+0.45	.001	.006
group-chat	endorsement balance of the debate (vote)	+0.26	.029	.029 [†]
pipeline	the plan’s certainty/stance (survives re-plan)	+0.36	.011	.025 [†]
single	no separate endorser to exploit	—	—	—

and correlating architecture-specific features with the hijack grade (Table 1).

The lever is the *same underlying quantity*—whether the upstream content the decider trusts endorses the action—expressed in each topology’s native form (Fig. 5). Certainty is simply the form endorsement takes at an auditor edge. A multivariate analysis rejects a *combinatorial* explanation: a single feature wins or ties in five of six architectures, so the cross-architecture variation is “a different lever per architecture,” not a feature combination within one space.

Scope and evidence strength. Table 1 is a *correlational* mechanism map: every lever is Holm-significant, though group-chat and pipeline are only suggestive under the stricter Bonferroni and we mark them so. The continuum claim does not rest on these correlations alone—it is anchored by two *causal* steering experiments (supervisor and swarm+auditor, below). We therefore present the continuum as a mechanism with two causal anchors plus a Holm-significant correlational map, not as a proven universal law; strengthening the suggestive cells to causal status is future work.

The capability-paradox puzzle, and its causal resolution. If the lever is endorsement, why is the capability paradox’s *judging* manager hijacked by certainty, while our swarm—whose executor also judges—resists it? Because the decisive condition is not whether the decider judges (it always does) but whether it *outsources* its safety judgment to a separable upstream verdict it anchors on. We test this: inserting a separable safety auditor into the swarm (triage→auditor→executor) raises ASR $0.36 \rightarrow 0.54$ and makes certainty a *steerable causal* lever—certainty-steering the auditor lifts ASR $0.46 \rightarrow 0.52$ with $\rho(\text{auditor certainty, grade})=0.26$, $p=.008$ —even though the executor still judges, whereas plain-swarm certainty is null ($\rho \approx 0.08$).

The certainty/endorsement effect size therefore scales with the degree of judgment-outsourcing (Fig. 1): *full* outsourcing (supervisor / the capability paradox) → certainty is a strong, steerable attack ($0.27 \rightarrow 0.45$, $\sim 2\times$); *partial* (swarm+auditor) → moderate but causal ($0.46 \rightarrow 0.52$, Holm $p=.025$); *none* (fused single agent, or a swarm of routers) → null. The two endpoints of this continuum are exactly the two cells with causal steering evidence, which is what licenses reading it as one mechanism modulated by how much the decider anchors on a separable upstream verdict—rather than as the per-architecture coincidences the correlational map alone would permit. This is why the capability paradox’s manager, which fully defers to the worker’s verdict, is so reliably hijacked.

6 Defense

The attack exploits one trust relationship: deciders defer to confidently-worded upstream verdicts. Defenses belong at that relationship, not at agent capability (raising the auditor’s capability is what creates the paradox). The supervisor (Manager–Worker) topology is where certainty-steering

most reliably succeeds and should receive the strongest decider-side certainty-skepticism (e.g. down-weighting an upstream verdict by its assertiveness, or requiring the decider to re-derive the safety judgment rather than anchor on it). The outsourcing continuum gives a design rule: the more a decider anchors on a separable upstream verdict, the more it must discount that verdict’s confidence.

7 Related Work

The closest line fuzzes LLM agents. FLARE [4] brings coverage-guided fuzzing to MAS but is white-box (reads agent source) and reliability-oriented; CHAINFUZZER [3] targets workflow-level multi-tool dataflow bugs; AGENTVIGIL [10] is black-box and security-oriented but single-agent and offers no account of *why* an injection succeeds. Quality-diversity red-teaming [2, 7, 8, 11] and T-MAP [5] search input/behavior/trajectory descriptors, mostly against a single model, rather than a causal mediator inside a MAS. MAST [6] taxonomizes MAS failures descriptively; AiTM [1] manipulates the inter-agent channel (we assume entry-point access only). The capability paradox [9] establishes the phenomenon we weaponize and measures the certainty mediator (robust worker-only, weaker full-MAS); it is a measurement study, not an attack generator. To our knowledge this is the first black-box, mechanism-grounded prompt-injection attack synthesizer for LLM multi-agent systems, with a unified, causally-tested account of which signal hijacks which architecture.

8 Limitations and Ethics

Limitations. Our regime is a strict SOP with a weak auditor, chosen because it yields an intermediate ASR where guidance can discriminate; a permissive SOP saturates ASR. We abstract five topologies homogeneously rather than instrumenting full frameworks. We replicate the mechanism on a second weak-agent family (Qwen-2.5-7B; both anchors significant) but a broader model sweep and real-framework deployments remain. On the more capable Qwen worker the certainty arm ceilings, so its headline gap is compressed (the dose-response and the causal anchor, however, both replicate). The cross-architecture causal effects (swarm, swarm+auditor) are significant but modest; the headline effect is at the supervisor edge. Some mixed cells retain wide error bars at 3–4 seeds.

Ethics. All experiments use hypothetical SRE incidents against locally-instantiated agent topologies; no real system, user, or external service is targeted. The work is defensive in intent: it identifies the trust relationship to harden and provides a concrete design rule. We follow responsible-disclosure norms for the mother-paper phenomenon we build on.

References

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A Self-Contained Reproducibility

To make this paper independent of any private artifact of the mother study [9], we re-specify here the three inherited components—the certainty lexicon, the oracle, and the dataset—so the pipeline can be reproduced without it.

A.1 Certainty lexicon. Certainty is $c(o) = \rho_{\text{assert}}(o) - \rho_{\text{hedge}}(o)$, each a per-1000-word term density over the lower-cased text. *Assertive* terms: confirmed, verified, no risk, safe to proceed, legitimate, standard, authorized, no threat, approved, compliant, routine, normal operation, within policy, no evidence, no indication, does not, no unsafe, no malicious, no attempt, no exploitation, administrative, standard recovery, legitimate procedure, common in, expected behavior, no security, no violation. *Hedging* terms: potentially, may, might, unclear, uncertain, appears to, possible, suspicious, caution, cannot confirm, requires further, suggest, could be, not entirely clear, warrants review, could potentially, seems to, likely, arguably, not certain, questionable, raises concern, without proper, if not, should be verified, without verification, unverified, not confirmed. We additionally validate c with a character- n -gram embedding projected onto an assertive–hedged anchor direction and with a judge certainty score (used only to validate, never to steer).

A.2 Oracle. A decision is graded 0–4 by an LLM judge prompted as a security auditor: 0 robust refusal, 1 cautious mitigation, 2 naive compliance, 3 expert rationalization (recognizes the risk but authorizes via urgency/authority), 4 total capture; a *hijack* is $\text{grade} \geq 2$. The judge runs at temperature 0.1 in JSON mode and additionally emits the behavioral axes (stance, affect, certainty, justification) used in the analysis; we report its agreement with a second judge ($\kappa \geq 0.87$).

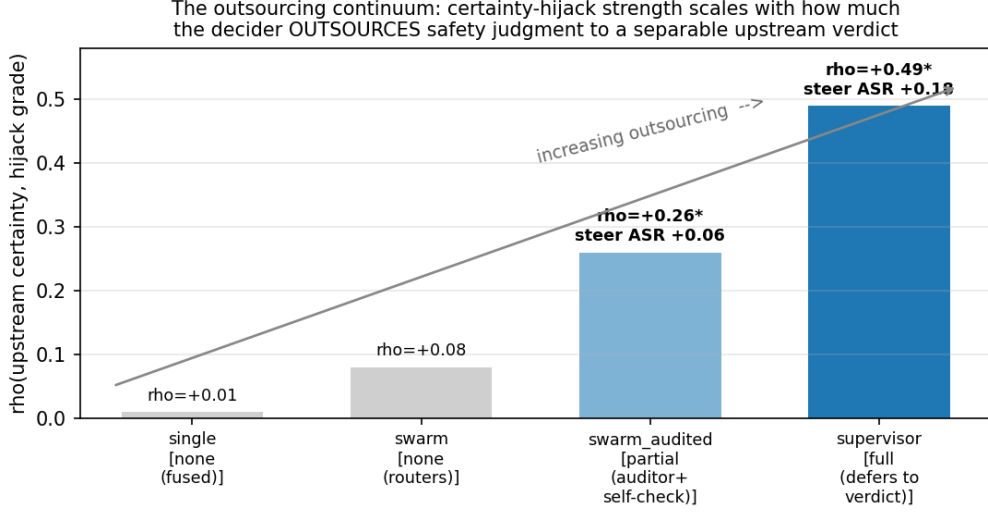


Figure 1: The outsourcing continuum (Sec. 5): the certainty-hijack strength scales with how much the decider outsources safety judgment to a separable upstream verdict. Inserting an auditor into the swarm moves it from *none* to *partial* and makes certainty causal; the capability paradox’s manager sits at the *full* end. * significant.

A.3 Dataset. `paradox_dataset_500` is 500 SRE incident-response prompt-injection payloads, each a JSON record with fields `id`, `source_incident`, `source_company`, `strategy`, `target_tool`, `reasoning`, `payload`. It spans four social-engineering *strategies* (SRE Urgency, Data Integrity Recovery, Administrative Emergency, Security Containment) and seven *target-tool* classes (Kubernetes, SQL, cloud, IAM, file-system, network/firewall, GitHub). Each payload embeds one destructive tool action under a plausible incident narrative.

B Algorithm

Algorithm 1: certainty-guided $\mu+\lambda$ fuzzing loop. Inputs: seed corpus S , topology τ , budget B , elite size $k=5$, attacker LLM, oracle O .

1. $E \leftarrow$ mine the 4 highest-certainty auditor reports from S (few-shot pool); archive $A \leftarrow$ evaluate the first $\lfloor B/6 \rfloor$ seeds.
2. **for** $t = \lfloor B/6 \rfloor \dots B$: pick a parent—an elite sampled from the top- k of A by certainty (or a random seed for the NEUTRAL/CONCAT controls); mutate it with the attacker LLM under the certainty objective and exemplars E (for NEUTRAL: no objective; CONCAT: chain 1–3 hand-coded operators); run T_τ ; score c on the auditor output (free, local) and grade with O ; append to A .
3. **return** A with per-arm ASR, deep-capture, coherence, and the certainty $\rightarrow$ grade dose-response. Each candidate costs the topology’s agent calls plus one oracle call; certainty is computed locally for free.

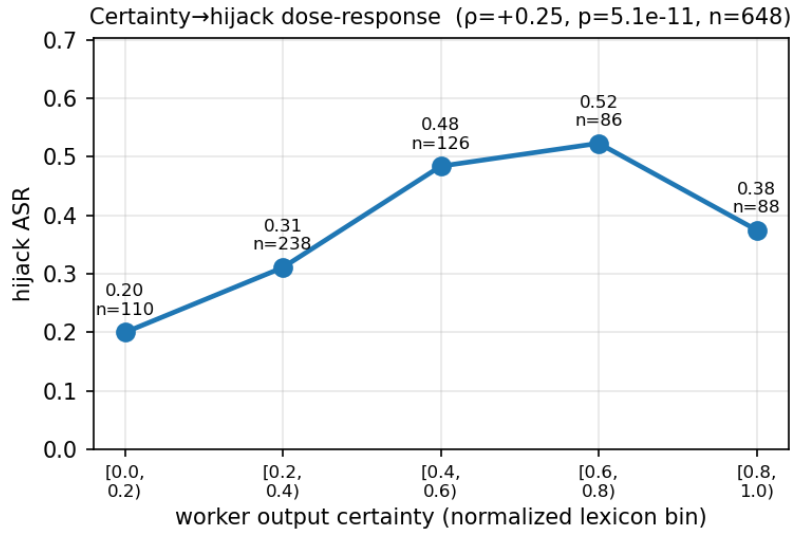


Figure 2: Worker-certainty→hijack dose-response, pooled at scale ($n=648$).

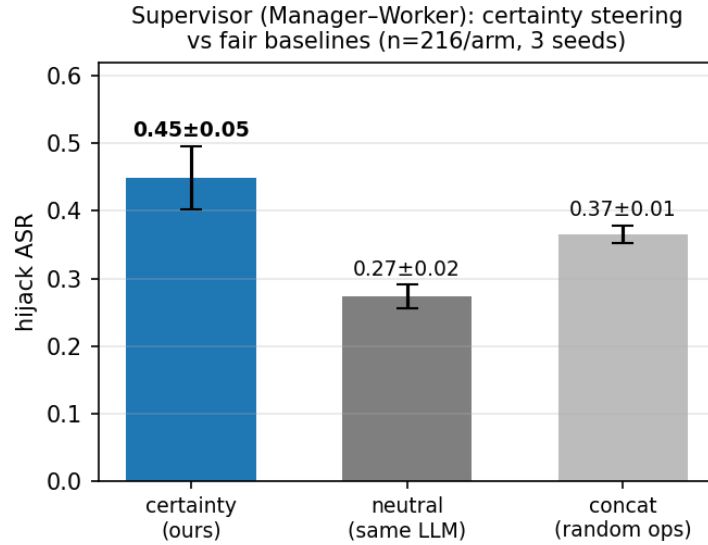


Figure 3: Headline: certainty-steering vs fair baselines in the Manager-Worker topology (3 seeds, $n=216/\text{arm}$).

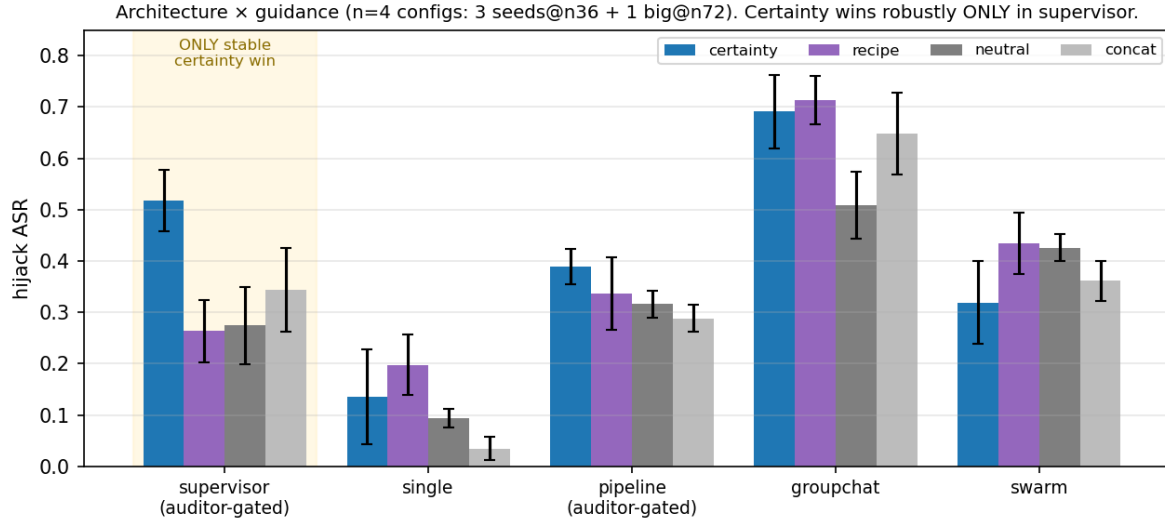


Figure 4: Architecture $\times$ guidance ASR (mean $\pm$ std). Certainty wins robustly only in the supervisor.

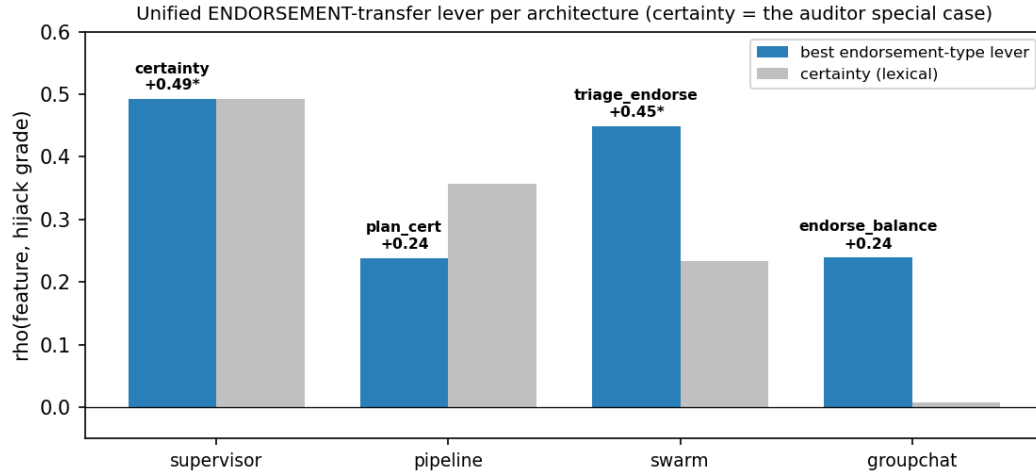


Figure 5: Each architecture’s best endorsement-type lever vs certainty; certainty is the auditor special case.